SIFS Workflow 2: From imzML to Spatially Informed Feature Selection

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Overview

This vignette starts from an imzML file and walks through the complete SIFS workflow:

1. Load and process the imzML file with `sifs_imzML_proc()`.
2. Select the matching sample from the label table and prepare the binary response.
3. Train an XGBoost classifier.
4. Compute SHAP values for the VT (viable tumor) class.
5. Compute binary HMCS values with the SPP/MPM/DSC spatial workflow.
6. Rank and visualize the m/z features.

The workflow uses a single label column, `Vt_vs_all_lbls`:

- `Vt_vs_all_lbls == "0"`: VT / tumor.
- `Vt_vs_all_lbls == "1"`: non-tumor.

No necrosis labels or other label columns are used in this vignette. For XGBoost and SHAP, the response is recoded so that **1 = tumor** and **0 = non-tumor**.

The binary HMCS comparison is therefore:

```
HMCS_VT = DSC(MPM hotspot, VT) - DSC(MPM hotspot, non-VT)
```

The non-VT reference is the complement of the VT mask. This is the Option A workflow: it avoids reading necrosis labels while preserving a binary VT-versus-reference HMCS contrast. One can use the same logic of binary label encoding for any other biological class(es).

```
library(SIFS)
```

Input data

The example uses three files:

- `positive-XIII-83318-recalibrated.imzML`
- `Lbls_df_Pos_all.rds`
- `fwhm_obj_pos.rds`

The dataset is shipped with the package. Set `SIFS_DATA_DIR` to the directory containing `imzML/`, `spectraLabels/`, and `fwhm_pos/`. If the variable is not set, the vignette checks `../dataset_for_SIFS` and `dataset_for_SIFS` relative to the working directory.

```
cat(
  "**External dataset not found.** The code chunks below are shown but not ",
  "evaluated. Set `SIFS_DATA_DIR` to the folder that contains ",
```

```
  "`imzML/`, `spectraLabels/`, and `fwhm_pos/` to run the full workflow."
)
```

1. Load and process the imzML file

The method named `sifs_imzML_proc()` performs the imzML-specific work in one place:

- reads the spectra,
- bins peaks using the supplied FWHM object,
- removes low-frequency peaks,
- creates the `sparseIntensityMatrix`,
- selects the requested/required Tissue sample from the labels, and
- aligns label rows to the sparse matrix rows.

The FWHM-derived binning tolerance is

```
FWHM at FWHM_setUp / FWHM_setUp
```

With `FWHM_setUp = 400` (can be changed to 200 for metabolites), this focuses the binning on the lipid region. The resulting tolerance is recorded in `obj$meta$fwhm_tolerance`.

```
obj <- sifs_imzML_proc(
  imzML_path = imzml_file,
  imzML_annotations_path = labels_file,
  fwhm_path = fwhm_file,
  labelType = "rds",
  sample_id = "t3",
  FWHM_setUp = 400,
  minFreq = 0.01
)
#> Input dataset is of type 'processed'.
```

The above snippet generates the `obj` object which contains a list of important objects required for the downstream processing, ML classification, SHAP calculation, and HMCS based feature ranking for the ROI.

```
str(obj, max.level = 1)
#> List of 7
#>  $ spmat :Formal class 'dgCMatrix' [package "Matrix"] with 6 slots
#>  $ spData:List of 3
#>   ..- attr(*, "class")= chr "sparseIntensityMatrix"
#>  $ coords:'data.frame':  3660 obs. of  2 variables:
#>  $ lbls  :'data.frame':  3660 obs. of  12 variables:
#>   ..- attr(*, "sifs_sample_id")= chr "t3"
#>   ..- attr(*, "sifs_label_rows_total")= int 255824
#>  $ spwin :List of 5
#>   ..- attr(*, "class")= chr "owin"
#>  $ mzList: num [1:8353] 100 100 100 100 100 ...
#>  $ meta  :List of 11
# Expected output:
# List of 7
#  $ spmat :Formal class 'dgCMatrix' [package "Matrix"] with 6 slots
#  $ spData:List of 3
```

```
#   ..- attr(*, "class")= chr "sparseIntensityMatrix"
#  $ coords:'data.frame':   3660 obs. of  2 variables:
#  $ lbls  :'data.frame':   3660 obs. of  12 variables:
#   ..- attr(*, "sifs_sample_id")= chr "t3"
#   ..- attr(*, "sifs_label_rows_total")= int 255824
#  $ spwin :List of 5
#   ..- attr(*, "class")= chr "owin"
#  $ mzList: num [1:8353] 100 100 100 100 100 ...
#  $ meta  :List of 11
```

```
dim(obj$spmat)
#> [1] 3660 8353
class(obj$spData)
#> [1] "sparseIntensityMatrix"
length(obj$mzList)
#> [1] 8353
nrow(obj$lbls)
#> [1] 3660
table(obj$lbls$pos_refNames)
#>
#>   t3
#> 3660
```

```
data.frame(
  sample = obj$meta$sample_id,
  spectra = obj$meta$n_spectra,
  features = obj$meta$n_features,
  fwhm_tolerance = signif(obj$meta$fwhm_tolerance, 6),
  minFreq = obj$meta$minFreq
)
#>   sample spectra features fwhm_tolerance minFreq
#> 1     t3    3660     8353     1.1759e-05    0.01
```

`obj$spData` is the `sparseIntensityMatrix` object with `spmat`, `mzAxis`, and `coordinates`. It is the object used by the spatial workflow; the vignette does not create a separate sparse matrix or load a precomputed SPMAT object.

The `spmat` element is the matrix used for XGBoost. The labels contain only the selected sample (`t3`) and are in the same row order as `spmat`.

```
stopifnot(
  nrow(obj$lbls) == nrow(obj$spmat),
  identical(obj$coords$x, obj$lbls$xCoord),
  identical(obj$coords$y, obj$lbls$yCoord),
  length(obj$mzList) == ncol(obj$spmat),
  inherits(obj$spData, "sparseIntensityMatrix")
)
```

2. Prepare the binary response

note:

Only `Vt_vs_all_lbls` is used. The raw encoding is 0 = tumor and 1 = non-tumor; The recoded response used by the model is 1 = tumor.

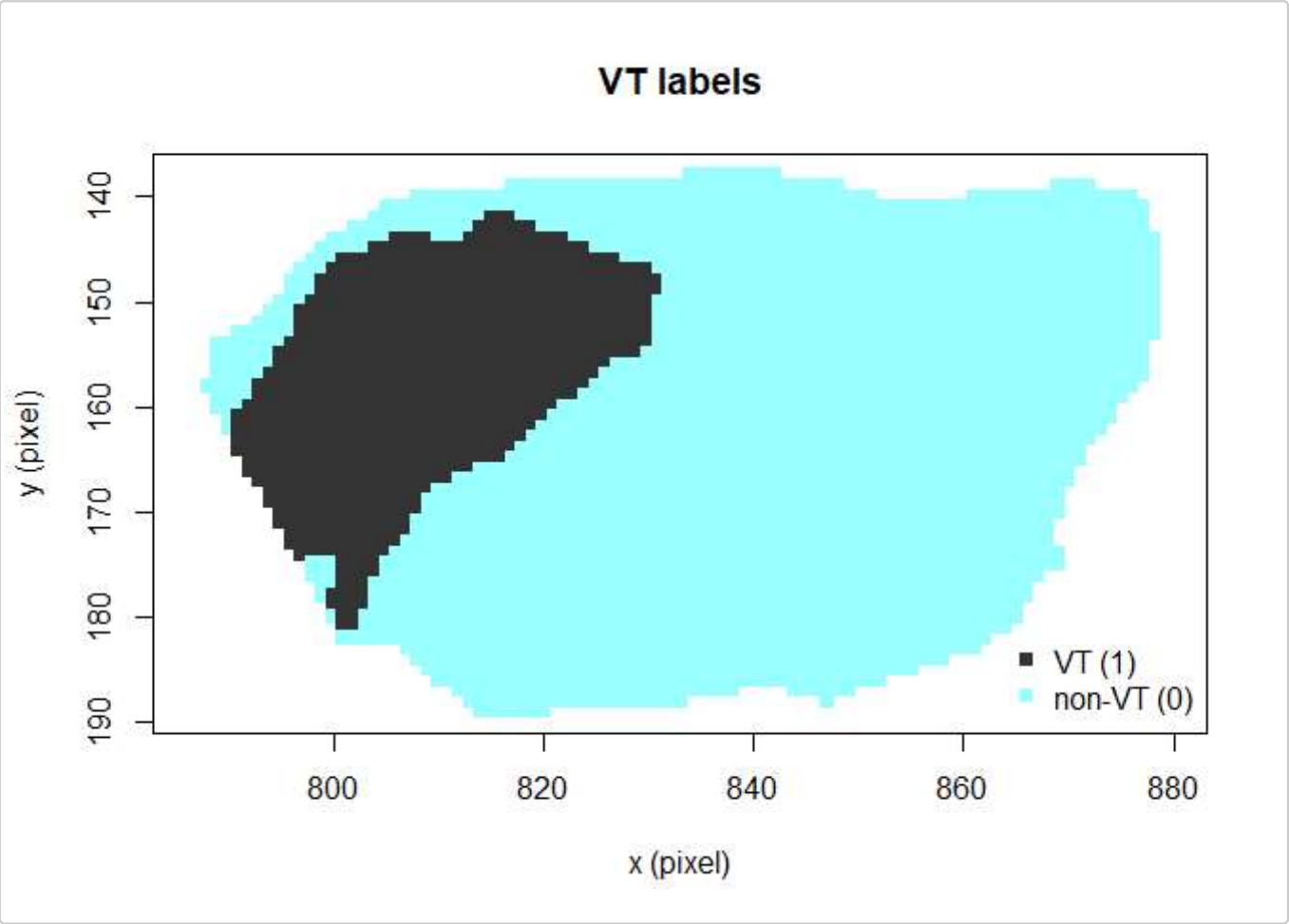
The positive class for most ML/DL classifiers are set to 1 by default, for that the following rectification is made.

```
lbls <- obj$lbls

y <- as.integer(as.character(lbls$Vt_vs_all_lbls) == "0")

table(y)
#> y
#>    0    1
#> 2837  823


blu = "#99ffff"
blk = "#333333"
plot(
  obj$coords$x,
  obj$coords$y,
  col = ifelse(y == 1L,blk, blu ),
  pch = 15,
  cex = 1.1,
  asp = 1,
  xlim = range(obj$coords$x),
  ylim = rev(range(obj$coords$y)),
  xlab = "x (pixel)",
  ylab = "y (pixel)",
  main = "VT labels"
)
legend(
  "bottomright",
  legend = c("VT (1)", "non-VT (0)"),
  col = c(blk, blu),
  pch = 15,
  bty = "n"
)
```



VT and non-VT pixels in the selected imzML section.

The model uses 3,660 pixels in total. In this section, 823 pixels are VT and 2,837 are non-VT. The `unwanted_spectra` column is intentionally not read.

3. Train XGBoost and compute SHAP values

`sifs_compute_shap()` trains the binary classifier and computes SHAP values for the VT pixels. The response passed to the function is the recoded `y`, so the function's `positive_class = 1` corresponds to tumor.

```
shap_results <- sifs_compute_shap(
  spmat_obj = obj$spData,
  labels = y,
  roi_name = "VT",
  train_ratio = 0.8,
  nrounds = 200L,
  positive_class = 1L,
  seed = 123L
)

head(shap_results$shap_df, 10)
#>       mzList   VT_SHAP
#> 1   770.5106 0.8274584
#> 2   885.6123 0.7049390
#> 3   771.5125 0.5015083
#> 4   863.6820 0.3487865
#> 5   977.7173 0.2981038
```

```
#> 6   732.5727 0.2548135
#> 7   617.1801 0.2319864
#> 8  1073.7336 0.1756635
#> 9   520.5087 0.1621455
#> 10  742.5721 0.1576287
```

The SHAP data frame has one row per m/z feature. It is ordered by the absolute mean SHAP value for the VT class.

4. Remove features with all-zero SHAP values

Most m/z features have no SHAP contribution. `sifs_filter_features()` removes features whose SHAP values are zero across all pixels.

```
filtered <- sifs_filter_features(
  shap_results = shap_results,
  focus_roi = "VT"
)

nrow(filtered$filtered_df)
#> [1] 416
filtered$removed_pct
#> [1] 95
```

The filtered data frame is used to choose the candidate m/z features for the spatial HMCS calculation. By default, the top 50 SHAP-ranked features are tested; this keeps the SPP/MPM/DSC step practical while preserving the strongest candidates.

5. Compute binary HMCS values

The spatial workflow uses `sifs_compute_hmcs()`:

1. `searchAnalyte()` builds an analyte point pattern for an m/z feature.
2. `probMap()` estimates a spatially significant hotspot map.
3. `dsc()` compares the hotspot window with the VT and non-VT windows.
4. `HMCS_VT = DSC_VT - DSC_nonVT`.

The MPM analysis is seeded (`seed = 123`) and uses a fixed bandwidth (`bw = 2.5`) so the ranking is reproducible.

```
fwhm_obj <- readRDS(fwhm_file)

vt <- as.character(lbls$Vt_vs_all_lbls) == "0"
nonvt <- !vt

mz_top <- head(filtered$filtered_df$mzList, top_n_hmcs)

hmcs_df <- sifs_compute_hmcs(
  spData = obj$spData,
  mz = mz_top,
  fwhm_obj = fwhm_obj,
```

```
  vt = vt,
  nonvt = nonvt,
  spwin = obj$spwin,
  bw = 2.5,
  seed = 123,
  min_points = 100,
  verbose = FALSE
)

# Confirm the final binary HMCS with the package calculator.
hmcs_df$HMCS_VT <- HMCS_calculatorBinary(
  hmcs_df,
  focusROI = "VT",
  referenceROI = "nonVT"
)

hmcs_df <- hmcs_df[order(-hmcs_df$HMCS_VT), , drop = FALSE]

knitr::kable(
  head(hmcs_df, 10),
  digits = 4,
  caption = "Top m/z features ranked by binary HMCS."
)
```

Top m/z features ranked by binary HMCS.

	mzRequested	mzList	n_spp	hotspot_area	dsc_mpm_VT	dsc_mpm_nonVT	HMCS_VT
29	755.5384	755.5384	3651	704	0.7911	0.0565	0.7346
33	733.5580	733.5580	3660	957	0.8360	0.1123	0.7237
6	732.5727	732.5727	3660	1041	0.8444	0.1310	0.7134
1	770.5106	770.5106	3316	973	0.8318	0.1186	0.7132
8	1073.7336	1073.7336	1721	940	0.8236	0.1133	0.7103
11	721.5564	721.5564	1107	1016	0.8363	0.1282	0.7081
28	720.5545	720.5545	3151	997	0.8286	0.1268	0.7018
44	790.5135	790.5135	2569	809	0.7892	0.0905	0.6987
5	977.7173	977.7173	968	963	0.8186	0.1221	0.6965
3	771.5125	771.5125	1606	1121	0.8405	0.1536	0.6869

Positive HMCS values mean that the MPM hotspot for a feature overlaps the VT region more than the non-VT region. Negative values indicate the opposite pattern.

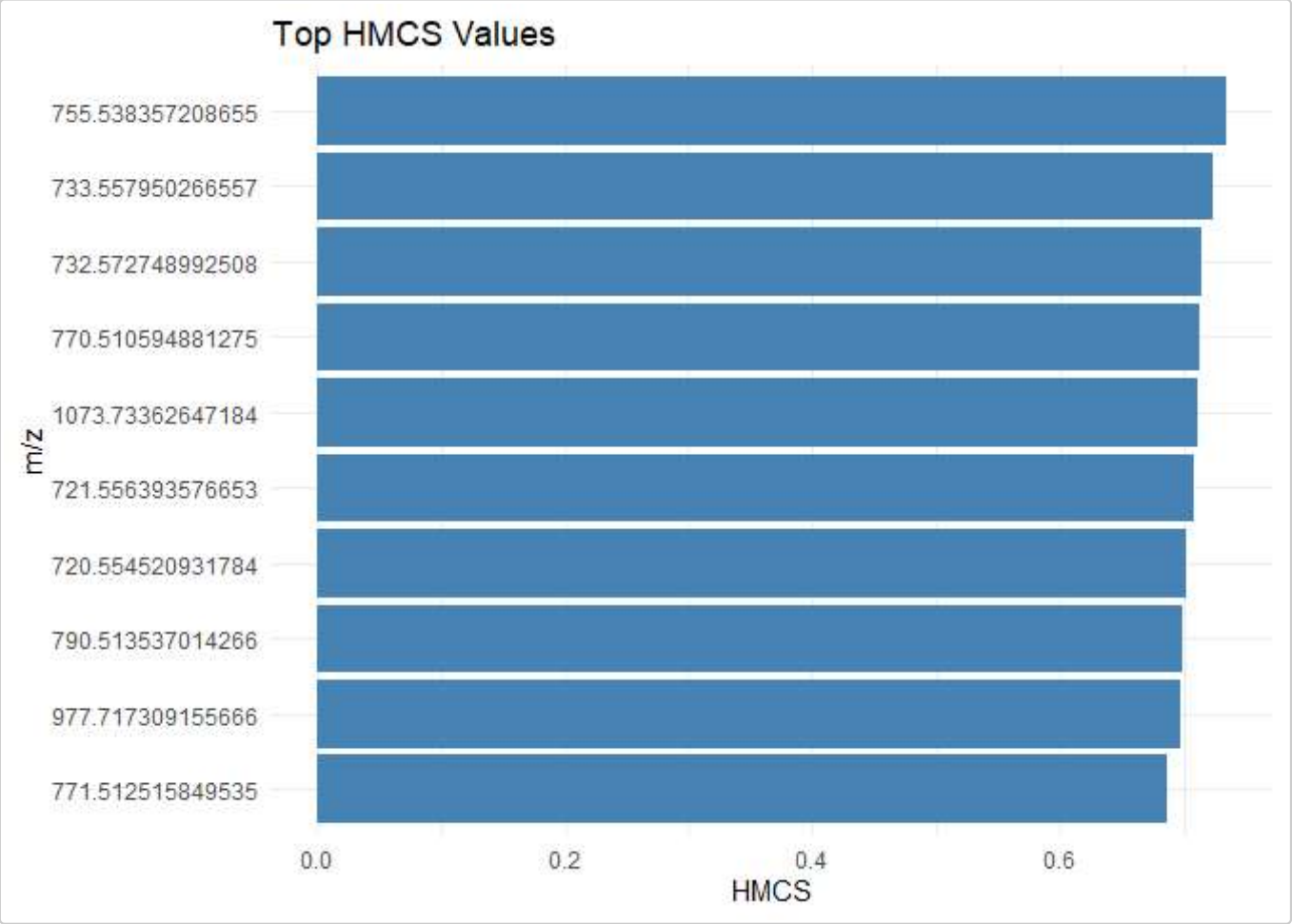
6. Visualize the rankings

```
hmcs_named <- stats::setNames(hmcs_df$HMCS_VT, hmcs_df$mzList)

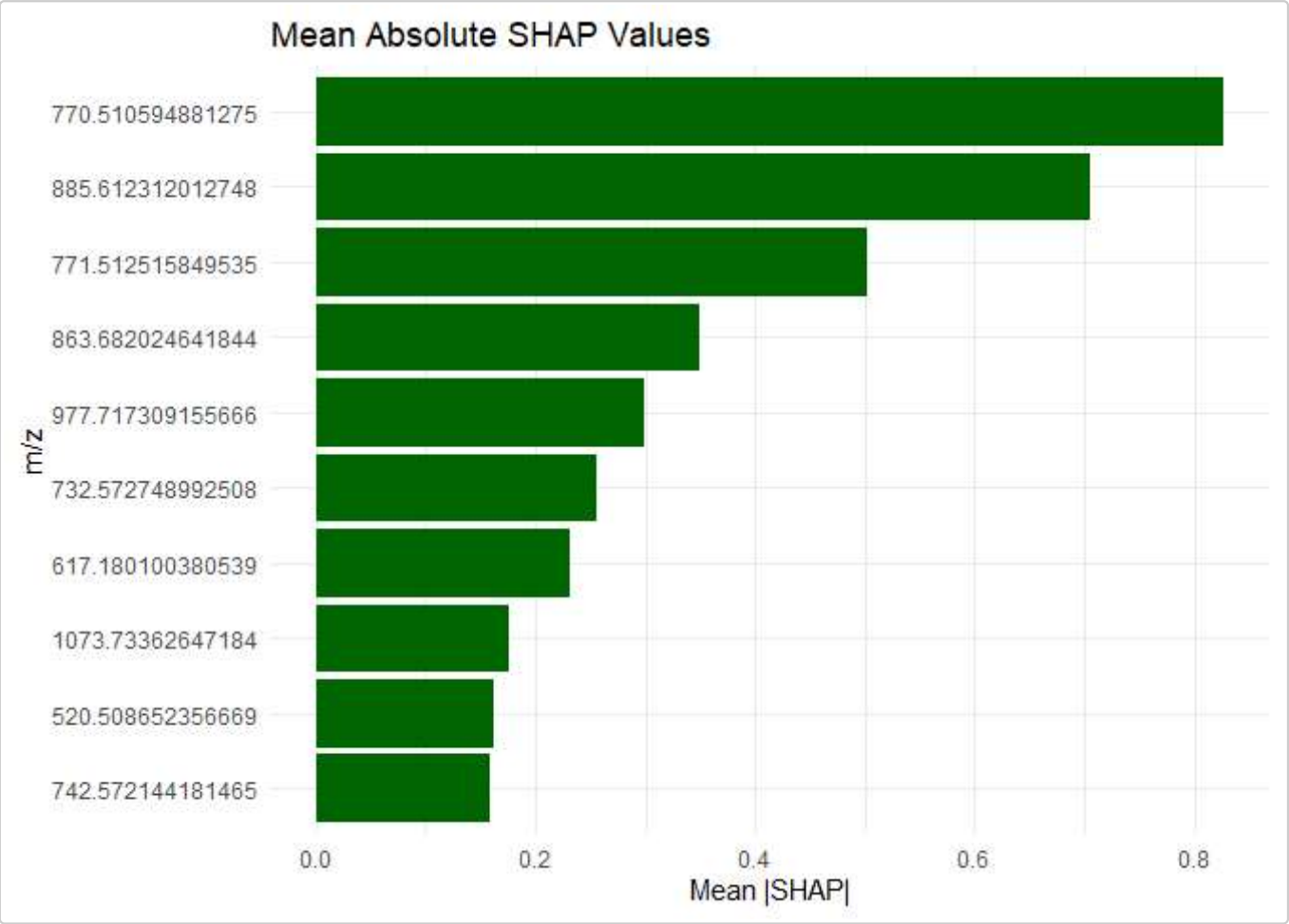
viz <- sifs_visualize(
  hmcs_results = hmcs_named,
  shap_df = filtered$filtered_df,
  output_dir = output_dir,
```

```
  top_n = top_n_plot
)
```

```
viz$hmcs_bar
```




```
viz$shap_bar
```




```
cat("Plots written to:", attr(viz, "output_dir"), "\n")
#> Plots written to: C:\Users\shada\AppData\Local\Temp\RtmpeWnv2K/SIFS_results
```

The HMCS bar plot shows the strongest positive spatial associations with VT. The SHAP bar plot shows the features with the largest mean absolute SHAP contribution. The two rankings answer related but different questions:

- **SHAP** asks which features the classifier uses most strongly.
- **HMCS** asks which features have hotspot maps that spatially resemble VT more than the non-VT reference.

7. Reproducibility notes

The example fixes the following settings:

- `sample_id = "t3"`
- `FWHM_setUp = 400`
- `minFreq = 0.01`
- `seed = 123` for the train/test split and model
- `seed = 123` for `probMap()`
- `bw = 2.5` for the MPM bandwidth
- `top_n_hmcs = 50` for the HMCS candidate set

The FWHM binning step is required. On this dataset, FWHM binning alone leaves more than 500,000 features; `minFreq = 0.01` reduces the axis to approximately 8,300 features, which is practical for XGBoost, treeshap, and the SPP/MPM/DSC loop.

Visualize MPMs for top n m/z

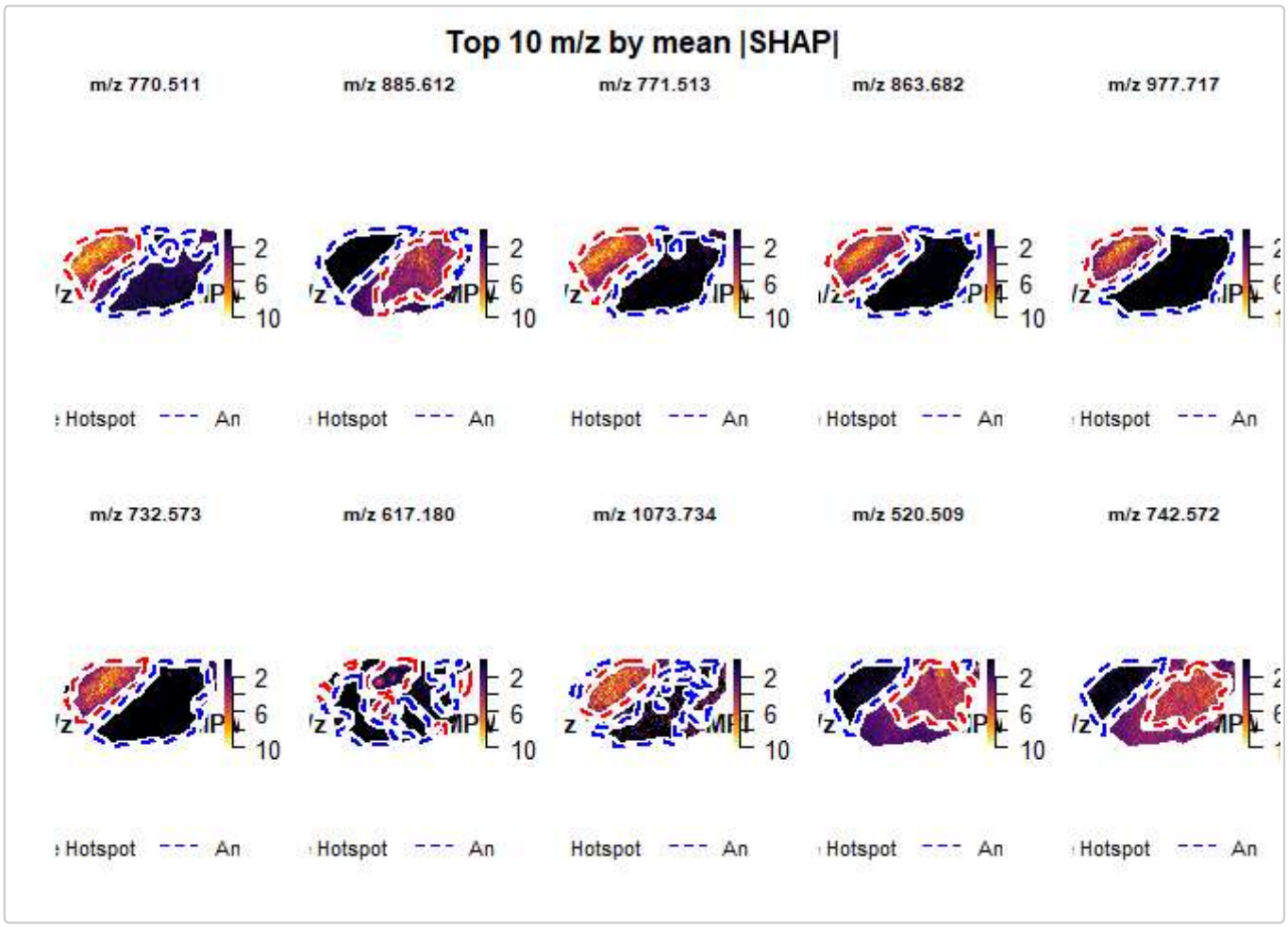
A. SHAP values

```
mz_top_10 <- shap_results$shap_df$mzList[1:10]

mpms <- lapply(mz_top_10, function(mz) {
  moleculaR::probMap(
    moleculaR::searchAnalyte(
      m            = mz,
      fwhm         = moleculaR::getFwhm(fwhm_obj, mz),
      spData       = obj$spData,
      spwin        = obj$spwin
    )
  )
})
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```

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#> done.
```

```
op <- par(mfrow = c(2, 5), mar = c(2, 2, 2, 1), oma = c(0, 0, 2, 0))
for (i in seq_along(mpms)) {
  moleculaR:::plot.molProbMap(mpms[[i]], what = "MPM")
  title(sprintf("m/z %.3f", mz_top_10[i]), cex.main = 0.9)
}
mtext("Top 10 m/z by mean |SHAP|", side = 3, outer = TRUE, line = 0, font = 2)
```




```
par(op)
```

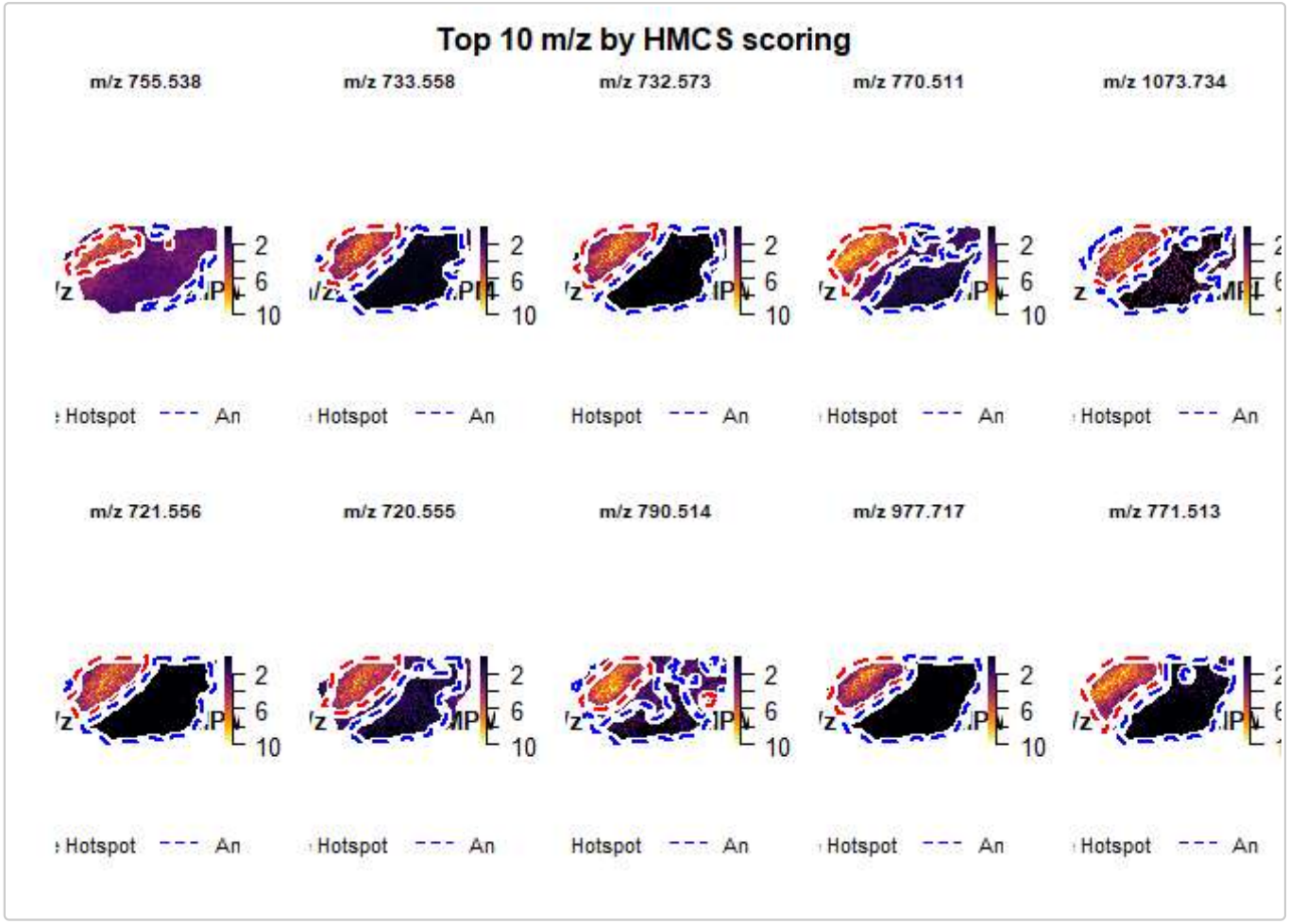
B. HMCS values

```
mz_top_10 <- hmcs_df$mzList[1:10]

mpms <- lapply(mz_top_10, function(mz) {
  moleculaR::probMap(
    moleculaR::searchAnalyte(
      m            = mz,
      fwhm         = moleculaR::getFwhm(fwhm_obj, mz),
      spData       = obj$spData,
      spwin        = obj$spwin
    )
  )
})
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#> done.
#>   |                                                                      |                                                                      |   0%  |                                                                      |==
        |  11%  |                                                                      |====                                                                  |  22%  |
        |=======                                                               |  33%  |                                                                      |=========                                                             |  44%  |
        |===========                                                           |  56%  |                                                                      |=============                                                         |  67%  |
        |================                                                      |  78%  |                                                                      |==================                                                    |  89%  |
        |====================| 100%
#> done.
#>   |                                                                      |                                                                      |   0%  |                                                                      |==
        |  11%  |                                                                      |====                                                                  |  22%  |
        |=======                                                               |  33%  |                                                                      |=========                                                             |  44%  |
        |===========                                                           |  56%  |                                                                      |=============                                                         |  67%  |
        |================                                                      |  78%  |                                                                      |==================                                                    |  89%  |
        |====================| 100%
#> done.
```

```
op <- par(mfrow = c(2, 5), mar = c(2, 2, 2, 1), oma = c(0, 0, 2, 0))
for (i in seq_along(mpms)) {
  moleculaR:::plot.molProbMap(mpms[[i]], what = "MPM")
  title(sprintf("m/z %.3f", mz_top_10[i]), cex.main = 0.9)
}
mtext("Top 10 m/z by HMCS scoring", side = 3, outer = TRUE, line = 0, font = 2)
```




```
par(op)
```

```
sessionInfo()
#> R version 4.6.1 (2026-06-24 ucrt)
#> Platform: x86_64-w64-mingw32/x64
#> Running under: Windows 11 x64 (build 26300)
#>
#> Matrix products: default
#>   LAPACK version 3.12.1
#>
#> locale:
#> [1] LC_COLLATE=English_United Kingdom.utf8
#> [2] LC_CTYPE=English_United Kingdom.utf8
#> [3] LC_MONETARY=English_United Kingdom.utf8
#> [4] LC_NUMERIC=C
#> [5] LC_TIME=English_United Kingdom.utf8
#>
#> time zone: Europe/Berlin
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] SIFS_0.0.0.9000
#>
#> loaded via a namespace (and not attached):
```

```
#>  [1] gtable_0.3.6             xfun_0.61                bslib_0.12.0
#>  [4] ggplot2_4.0.3            raster_3.6-32            readBrukerFlexData_1.9.3
#>  [7] spatstat.sparse_3.2-0    lattice_0.22-9           vctrs_0.7.3
#> [10] tools_4.6.1              spatstat.utils_3.2-5     generics_0.1.4
#> [13] goftest_1.2-3            parallel_4.6.1           treeshap_0.4.0
#> [16] tibble_3.3.1             readMzXmlData_2.8.4      pkgconfig_2.0.3
#> [19] Matrix_1.7-5             data.table_1.18.6.1      RColorBrewer_1.1-3
#> [22] S7_0.2.2                 lifecycle_1.0.5          compiler_4.6.1
#> [25] farver_2.1.2             deldir_2.0-4             textshaping_1.0.5
#> [28] terra_1.9-50             moleculaR_0.9.5          codetools_0.2-20
#> [31] spatstat.explore_3.8-3   htmltools_0.5.9          sass_0.4.10
#> [34] yaml_2.3.12              pillar_1.11.1            jquerylib_0.1.4
#> [37] cachem_1.1.0             spatstat.univar_3.2-0    viridis_0.6.5
#> [40] abind_1.4-8              MALDIquant_1.22.3        nlme_3.1-169
#> [43] spatstat.geom_3.8-3      tidyselect_1.2.1         digest_0.6.39
#> [46] dplyr_1.2.1              labeling_0.4.3           polyclip_1.10-7
#> [49] fastmap_1.2.0            grid_4.6.1               cli_3.6.6
#> [52] MALDIquantForeign_0.14.1 magrittr_2.0.5           base64enc_0.1-6
#> [55] XML_3.99-0.25            spatstat.data_3.1-9      withr_3.0.3
#> [58] tensor_1.5.1             scales_1.4.0             sp_2.2-4
#> [61] xgboost_3.2.1.1          rmarkdown_2.32           otel_0.2.0
#> [64] gridExtra_2.3.1          ragg_1.5.2               evaluate_1.0.5
#> [67] knitr_1.52               viridisLite_0.4.3        rlang_1.3.0
#> [70] Rcpp_1.1.2               spatstat.random_3.5-2    glue_1.8.1
#> [73] rstudioapi_0.19.0        jsonlite_2.0.0           R6_2.6.1
#> [76] systemfonts_1.3.2
```